

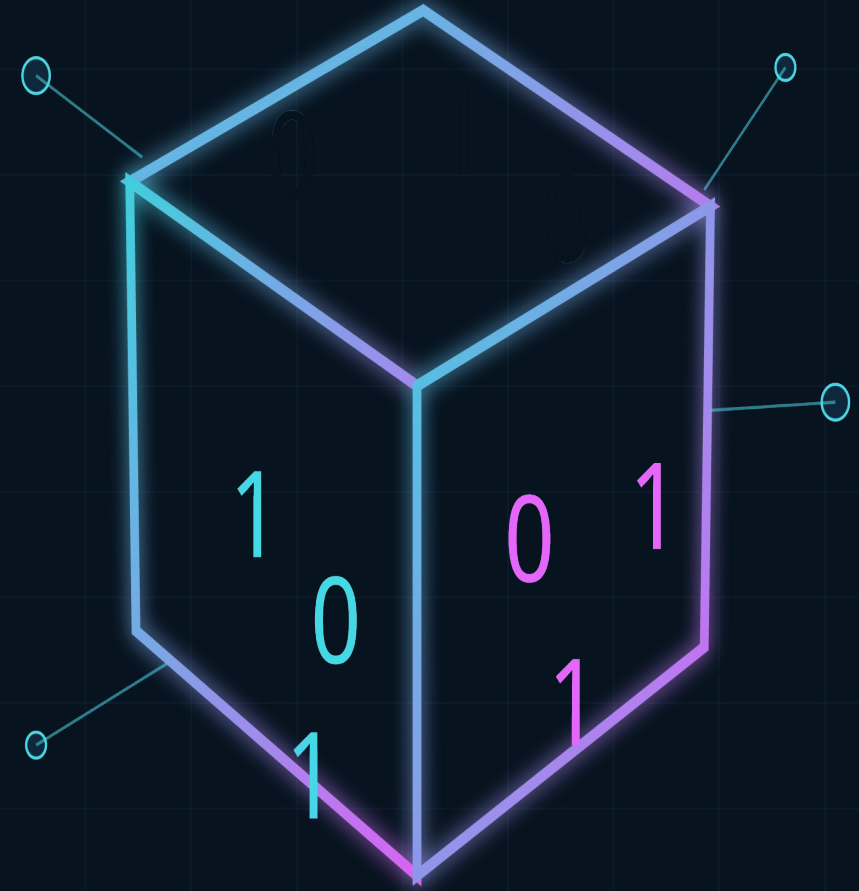
EVERYTHING IS BITS

Bits, Bytes, and Digital Information

30 min instruction • 30 min guided exercise

PYTHON 101 • DAY 1 AM

The computer stores patterns. Software supplies the meaning.



Cold open: What is 01000001?

The bits are real. The meaning is a choice.

0 1 0 0 0 0 0 1

65

a decimal number

A

an ASCII character

65

a dark-gray pixel value

...

part of an instruction

“Which answer is correct?”

Answer: potentially all of them — context decides.

Today's mission

Peel back one layer beneath files, apps, and screens.

- 1 Build** numbers from binary place values
- 2 Explain** bits, bytes, and common size units
- 3 See** how text, images, and audio become numbers
- 4 Recognize** that the same bits can mean different things

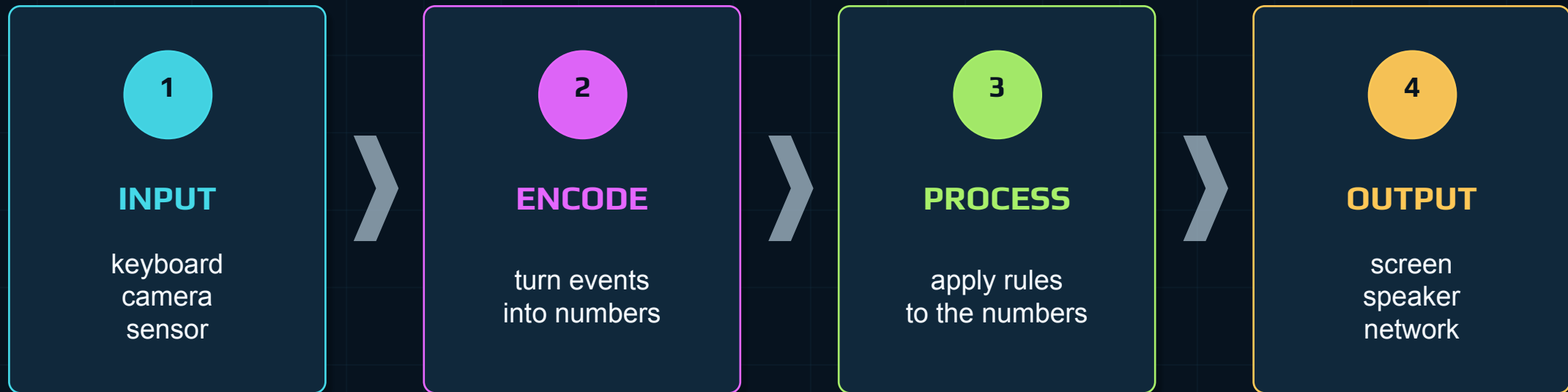
ONE IDEA TO KEEP

**Bits have
no built-in
meaning.**

A format, type, program, or
protocol tells the computer how
to interpret them.

Computers manipulate symbols

They follow rules extremely quickly; they do not “see” meaning the way humans do.



Human meaning

Machine representation

The “meaning” is preserved only when both sides agree on the encoding rules.

A bit: one binary choice

Binary means there are two possible states.

0

off • false • no

1

on • true • yes

A physical device does not have to look like a switch.

It only needs two reliably distinguishable states.

BIT

binary digit

**The smallest common unit
of digital information**

More bits = more possible patterns

With n bits, there are 2^n possible combinations.

1 bit**0 / 1****2 values****2 bits****00 01 10 11****4 values****4 bits****0000 ... 1111****16 values****8 bits****00000000 ... 11111111****256 values**

Doubling the number of patterns does not require twice as many bits — it requires only one more bit.

A byte is eight bits

Eight bits can form 256 distinct patterns: 0 through 255 when interpreted as an unsigned whole number.



1 byte

8 bits

0

smallest unsigned value

182

the shown bit pattern

255

largest unsigned value

From bytes to terabytes

Storage sizes are counts of bytes. Real files vary — these examples are deliberately approximate.

B**BYTE**
8 bits

one ASCII character

KB**KILOBYTE**
≈ 1,000 B

a short plain-text page

MB**MEGABYTE**
≈ 1,000 KB

about 1 min of MP3 audio

GB**GIGABYTE**
≈ 1,000 MB

hundreds of photos or songs

TB**TERABYTE**
≈ 1,000 GB

a modern large drive

Note: You may also see KiB, MiB, and GiB for exact powers of 1,024.

Decimal and binary use the same idea

Only the base — and therefore the place values — changes.

BASE 10**4 3 7**

100s 10s 1s

$$4 \times 100 + 3 \times 10 + 7 \times 1 = 437$$

BASE 2**1 1 0 1**

8s 4s 2s 1s

$$1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1 = 13$$

Rightmost place is always the “ones” place. Move left and multiply by the base.

Convert 13 to binary

Find which powers of two add up to the decimal number.

8

USE IT

1

4

USE IT

1

2

SKIP IT

0

1

USE IT

1

$$13 = 8 + 4 + 1$$

1

1

0

1

$$13_{10} = 1101_2$$

Text is stored as numeric character codes

A character encoding is an agreement between numbers and symbols.

P

80

Y

89

T

84

H

72

O

79

N

78

ASCII

An early, widely used encoding for English letters, digits, punctuation, and control codes.

Encoding turns “PYTHON” into numbers. Decoding turns the numbers back into “PYTHON.”

Unicode makes room for the world

Modern text needs far more than the original English-focused ASCII set.

A

Latin letter

U+0041

é

accented letter

U+00E9

你

Chinese character

U+4F60

Ω

Greek letter

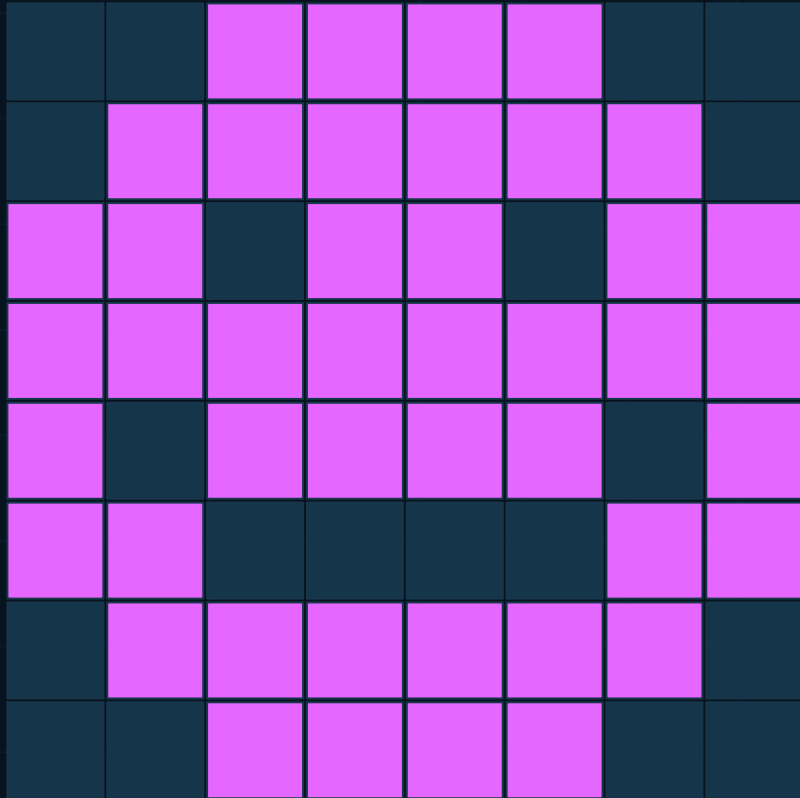
U+03A9

Unicode defines characters. UTF-8, UTF-16, and others define how those characters become bytes.

A character is not always one byte.

Images are grids of numeric pixel values

A pixel stores numbers that describe color and brightness.



8 × 8 pixels

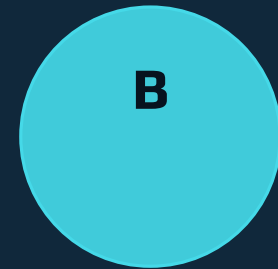
One common color model: RGB



255



140

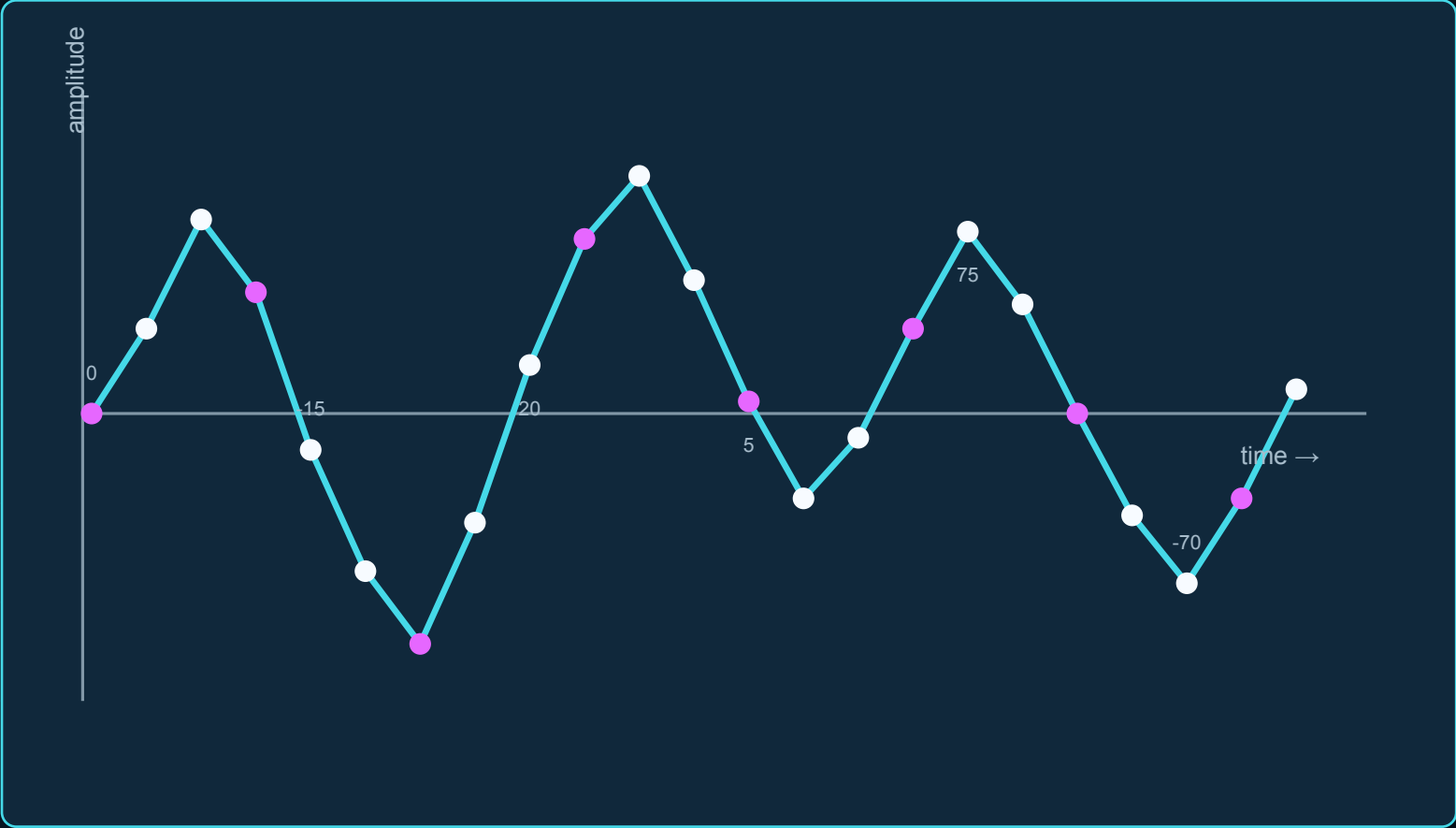


40

Compression stores the same visual information more efficiently — sometimes perfectly, sometimes with controlled loss.

Audio is a sequence of numeric samples

A microphone's changing signal is measured repeatedly over time.



More samples

capture changes
more frequently

More bits per sample

allow more possible
amplitude values

Compressed audio reduces how many bytes must be stored or streamed.

The same bits need the right decoder

Correct interpretation depends on context and agreed rules.

0

1

0

0

0

0

0

1

NUMBER

65

unsigned integer

TEXT

A

ASCII / Unicode

IMAGE

65

pixel intensity

CODE

...

architecture + position

KEY CONCEPT: Bits do not inherently represent text, numbers, pictures, sound, or instructions.

GUIDED EXERCISE

Decode the digital world

Work in pairs or small groups. Explain your reasoning — not just the final answer.

01 Binary

02 Patterns

03 Text

04 Files

ABOUT 30 MINUTES

Round 1: Binary and possible values

Show your work using powers of two.

A • Convert decimal to binary

1 $5_{10} =$ _____

2 $10_{10} =$ _____

3 $13_{10} =$ _____

4 $18_{10} =$ _____

B • How many possible values?

1 1 bit $2^n =$ _____

2 2 bits $2^n =$ _____

3 4 bits $2^n =$ _____

4 8 bits $2^n =$ _____

Bonus: What is the largest unsigned whole number that fits in 4 bits?

Round 1 answers

Check the method, not only the pattern.

$5_{10} \rightarrow 0101_2$ $4 + 1$

1 bit $2^n =$ 2 values

$10_{10} \rightarrow 1010_2$ $8 + 2$

2 bits $2^n =$ 4 values

$13_{10} \rightarrow 1101_2$ $8 + 4 + 1$

4 bits $2^n =$ 16 values

$18_{10} \rightarrow 10010_2$ $16 + 2$

8 bits $2^n =$ 256 values

Bonus: $1111_2 = 15_{10}$, so 4 bits store unsigned values 0–15.

Round 2: Decode the message

6 MINUTES

Use the mini ASCII reference.

80 89 84 72 79 78

Decoded text: _____

Then answer:

1. What must sender and receiver agree on?
2. Would these numbers always mean text? Why not?

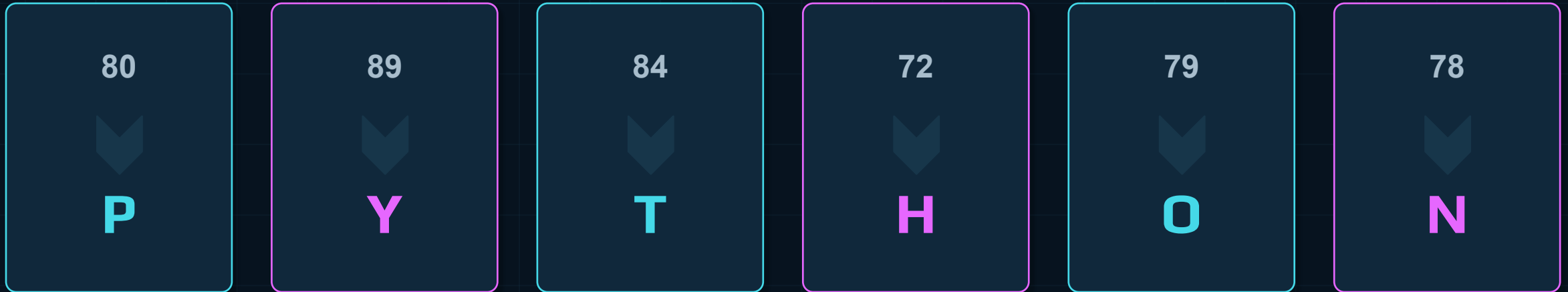
MINI REFERENCE

A	65
H	72
N	78
O	79
P	80
T	84
Y	89

Hint: map each number independently.

Round 2 answer

The numbers become text only because we chose a character encoding.



PYTHON

Agreement required: encoding + order + boundaries. Without context, the values are only values.

Round 3: File sizes and file types

Use reasoning and rough scale — exact sizes vary.

A • Order from smallest to largest

A	Plain-text notes	20 KB
B	Phone photograph	4 MB
C	Four-minute song	7 MB
D	Five-minute HD video	500 MB

Explain why video is usually much larger than text.

B • What does the extension help with?

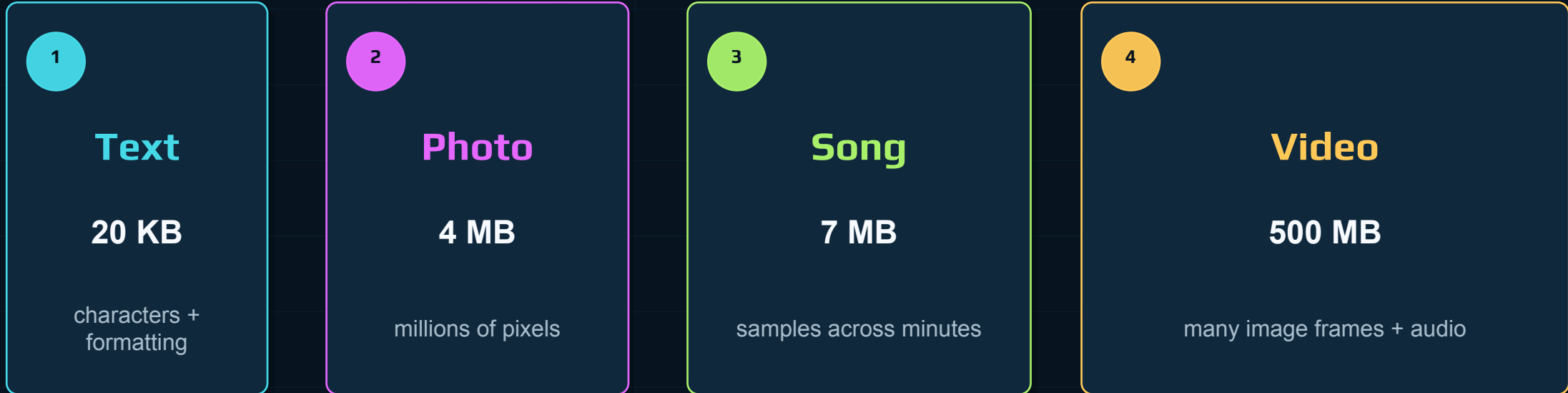
report.txt	photo.jpg
song.mp3	archive.zip

Discuss:

- What does software need to know?
- Does renaming photo.jpg to photo.txt convert it?
- What else could identify the format?

Round 3 discussion

Storage follows the amount of information and the chosen encoding/compression.



A file extension is a hint, not a transformation.

Software may also inspect headers/signatures, metadata, and internal structure before choosing a decoder.

Renaming cat.jpg → cat.txt changes the label — not the bytes inside.

The five ideas to leave with

Everything else in computing builds on these.

01 Bits Two possible states: 0 or 1.

02 Bytes Eight bits; 256 possible patterns.

03 Binary Place values are powers of two.

04 Encoding Text, pixels, and samples become numbers.

05 Interpretation Context tells software what the bits mean.

EXIT TICKET: In one sentence, explain why the same byte sequence can mean different things.